

Choose a safe sampling rate

Background

A sum of sinusoids is band-limited by its highest component frequency $f_{\max}$. Its Nyquist rate is $2 f_{\max}$. Use the lecture's strict condition $f_s > 2 f_{\max}$ for guaranteed recovery. Equality is a boundary case, not a safe result in this task.

Your task

For each candidate sampling rate, report `SAFE`, `BOUNDARY`, or `ALIAS_RISK`. Also report the Nyquist rate. Display the supplied comparison plot. Classification concerns the recovery guarantee, not whether every possible waveform necessarily fails.

Function contract and assumptions

`assess_rates(freqs, rates)` returns `(nyquist, labels)`. `freqs` is a nonempty array of positive component frequencies in Hz; all components have nonzero amplitude. `rates` contains positive Hz values. `labels` is a list in the same order as `rates`. Test data use exact integer frequencies and rates.

Example / correctness check

`freqs = [3, 7, 12]`, `rates = [20, 24, 30, 48]` gives `nyquist = 24` and `labels = [ALIAS_RISK, BOUNDARY, SAFE, SAFE]`.

Exam instructions

Complete only the `TODO` functions in `template.py` and preserve their signatures. The supplied driver handles printing and plotting. Your functions must work for all valid inputs described above, not just the example. No external files or packages beyond NumPy and Matplotlib are needed. Small floating-point differences are acceptable.

Topic reference: PDF pages 19-22.